

Henry Routson

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Hi, I'm Henry. I have extensive experience knowledge and experience in relation to modern AI systems and Software Engineering.
I am currently working as an AI Engineer.

Education

The University of Melbourne

The University of Melbourne is considered to be the most competitive and academically challenging university in the country.
Over my time I did work for top clubs, created numerous personal projects, and developed high-level software engineering skills.

Bachelor of Science, Major in Computing and Software Systems - distinction
Master of Computer Science, Major in AI - (in progress, part time)
NLP - Fine Tuning Models, Prompt tuning, Transformers, Hugging-face

Ballarat Grammar

Academic scholarship
VCE: Software Development, Specialist and Methods Mathematics, Chemistry and Physics

Experience

KPMG, AI Engineer (2025 June - Present) - <https://kpmg.com/au/en.html>

AI Prototype Engineer in Futures. Working with LLMs and agent frameworks.

Developed standardised high-level codebase (34 thousand lines) for all AI developers to automate prompting, parsing, validation, and deep research, resulting in accelerated development.
Facilitated the automated assessment of 10k pages of legislation equivalent.
Created pipeline to automate SQL migration and database merging.
Deployed local Transformer models to CUDA GPUs with Hugging face and Ollama libraries including LLMs and OCR.
Developed autonomous agent system with RLMS (Recursive language models)
Multi agent and sub agent implementation.

Skills : Azure cloud, Cloud deployment, FastAPI REST API development, LangExtract, LangChain, Pandas, Production deployments, docker.

Bet Right, Data Science Intern (2025 Jan - April) - <https://www.linkedin.com/company/bet-right/>

Working with a broad range of tools including SQL, Snowflake, Databricks, PowerBI, Azure and DevOps.
Headed improved abuse detection ML model to improve performance by between 40-15% with stacking.

DSCubed, DS AI team - vice head (Nov 2024 > 2025 July) - <https://www.dscubed.org.au/>

Working on Retrieval Augmented Generation, Vector databases, LLM frameworks and APIs
Contributing to leading open source Github projects like llama index.

Aerospace and Rocket Engineering Society, The University of Melbourne - (2023) - <https://www.linkedin.com/company/aresunimelb>

ARES is the rocket society at the University of Melbourne.
Each year we launch a 2 meter tall composite rocket with the goal of hitting as close to 30k feet as possible.
My role was to develop 3D front end software to assist engineers in simulating the trajectory of the rocket across supersonic and subsonic speeds.

Leetcode competitive programming : <https://leetcode.com/u/HenryRoutson/>

Open-source contribution

Cached File Explorer - in rust : <https://github.com/conaticus/FileExplorer/pull/47>

· This was a project to create an ultra fast file explorer using caching and the low-level rust language.

- I fixed poisoned thread issue which can arise when using multi-threading.

Supabase SQL Auth UI <https://github.com/supabase-community/flutter-auth-ui/pull/108>

- Supabase is a popular PostgreSQL company which handles application authentication and tooling.
- I added additional configuration options and added documentation to the authentication UI for use in my personal project.

Llama index : https://github.com/run-llama/llama_index/pull/17393

- In relation to my AI engineer role, for data intake we required data from notion for which a quality API wrapper did not exist.
- I create a merge to fix multiple bugs, add new features and improve code quality in the notion reader.

Academics

Machine learning (Keras, Scikit-learn)

- Created a stacked ML model including a neural network.

AI (Python)

- Implemented Monte Carlo tree search.

IT Project (JavaScript / TypeScript, React, HTML, CSS)

- Increased code quality and redesigned stack visualisation for a Quicksort visualisation.
- **Try Here!** <https://dev-aia.vercel.app/>

Computer Systems (C)

- Functional memory allocator for operating systems and multi threaded web server.

Algorithms and Data Structures (C)

- Implemented a Quad-tree, Linked list, Dijkstra and A*.

Foundations of Computing (Python)

- Achieved overall 98%, one of the highest marks in the cohort.

Personal projects

https://github.com/HenryRoutson/llm_experiments

- This is my repo where I experiment with transformers, JAX, Keras, PyTorch and LLM repositories.
- One of my experiments involves attempting to combine pre-training and supervised fine tuning.

https://github.com/HenryRoutson/Auto_Avail/blob/main/images/groups.png (JS / TS)

- This is a version of <https://www.when2meet.com/>.
- Uses location data and preferences to generate continuous periods.
- Utilises Supabase, previously Firebase, Flutter, React native, SQL / Postgres, pgTAP, user authentication, Expo, Typescript, cloud functions, Cursor IDE.

<https://github.com/HenryRoutson/CHelp> (C)

- When learning C in university I was frustrated when trying to find bugs with memory debuggers like Valgrind.
- Created a valgrind alternative. Tracks dynamic allocations in C and catches issues early to assist debugging.

<https://github.com/HenryRoutson/autoheader> (Rust)

- Generates header files for C automatically.

References

(Please contact me to request references)

Cindy is an AI engineer at KPMG I worked closely with across my time there. <https://www.linkedin.com/in/runxin-liu/>

Pranav is a level 3 Data Scientist at NAB. <https://www.linkedin.com/in/pranavjayanty/>